

# Preference-Aware Bayesian Optimization for Interactive Decision Making

Yujie Ma, Ludi Wang, Wenjuan Cui, Yuanchun Zhou<sup>✉</sup>, *Member, IEEE*, and Yi Du<sup>✉</sup>, *Member, IEEE*

**Abstract**—In real-world scenarios, optimization problems generally exhibit multiobjective characteristics, necessitating a balance among conflicting evaluation criteria. An effective approach to multiobjective optimization (MOO) is to approximate the Pareto front. However, completely solving for the Pareto front incurs extremely high computational costs, making it almost infeasible in practical applications. In practical scenarios, decision makers (DMs) typically require only a single most preferred solution from the Pareto-optimal set. Existing optimization methods often struggle to incorporate real-time DMs' preferences during the optimization process. To address this limitation, this article proposes a preference-aware Bayesian optimization (PABO) framework for interactive decision-making that seamlessly integrates DMs' feedback throughout the entire optimization process. By embedding preference information into the candidate solution generation stage, PABO dynamically adjusts the balance between exploration of uncertain regions and exploitation of preference-aligned solutions, thereby achieving efficient preference satisfaction. Experiments on benchmark functions and real-world engineering cases demonstrate that PABO achieves comparable or superior solution quality with significantly fewer expensive evaluations than state-of-the-art methods. This achievement indicates that PABO demonstrates significant advantages in improving optimization efficiency and reducing costs, providing a more feasible technical approach for the practical application of MOO problems.

**Index Terms**—Bayesian optimization (BO), expensive problems, multiobjective optimization (MOO), user-preference incorporation.

## I. INTRODUCTION

**R**EAL-WORLD optimization problems often involve multiple, conflicting objectives [1]—for example,

Received 18 September 2025; revised 31 October 2025; accepted 10 November 2025. Date of publication 25 November 2025; date of current version 18 March 2026. This work was supported in part by the Natural Science Foundation of China under Grant T2322027 and Grant 62442204, in part by the National Key Research and Development Plan of China under Grant 2022YFF0712200, in part by the Youth Innovation Promotion Association of CAS, and in part by the Key Research Program of Chinese Academy of Sciences under Grant RCJJ-145-24-20. This article was recommended by Associate Editor H. Gao. (Yujie Ma and Ludi Wang contributed equally to this work.) (Corresponding author: Yi Du.)

Yujie Ma, Ludi Wang, and Wenjuan Cui are with the Computer Network Information Center, Chinese Academy of Sciences, Beijing 100083, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: yjma@cnic.cn; wld@cnic.cn; wenjuancui@cnic.cn).

Yuanchun Zhou and Yi Du are with the Computer Network Information Center, Chinese Academy of Sciences, Beijing 100083, China, also with Hangzhou Institute for Advanced Study, UCAS, Hangzhou 310000, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: zyc@cnic.cn; duy@cnic.cn).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCYB.2025.3633283>.

Digital Object Identifier 10.1109/TCYB.2025.3633283

maximizing efficacy while minimizing cost in pharmaceutical development, or balancing risk and return in financial portfolio design. Bayesian optimization (BO) [2] and data-driven evolutionary optimization [3], [4], [5], [6] are two extensively adopted techniques for tackling such expensive optimization problems. BO specifically has become a powerful tool for tackling functions with unknown internal structure by building probabilistic surrogates and guiding the search with acquisition functions. It has demonstrated remarkable efficiency in domains ranging from hyperparameter tuning of machine-learning models [7], [8], [9], [10], [11] to engineering design automation [12], [13], [14], [15] and financial portfolio optimization [16], [17]. However, generating the complete Pareto front incurs prohibitive computational costs when function evaluations are expensive. Existing multiobjective BO methods lack mechanisms to adaptively integrate real-time decision-maker (DM) preferences, often expending computational resources on regions of the Pareto front that mismatch practical requirements. More critically, real-world decision-making scenarios seldom necessitate exhaustive exploration of all optimal solutions. Instead, DM typically seeks a single preferred solution aligned with their priorities. These challenges underscore the urgent need for frameworks that seamlessly integrate preference learning with multiobjective optimization (MOO), enabling adaptive alignment with DM priorities while maintaining computational tractability.

Despite some efforts to incorporate DM preferences, current preference-aware MOO methods still exhibit significant drawbacks. Approaches relying on static prior weight assumptions require DM to explicitly specify reference points or importance rankings of objectives before optimization begins [18], [19], [20], [21]. While this approach simplifies the problem by reducing computational complexity and establishing explicit alignment with initial priorities in the objective space, it fundamentally relies on two critical assumptions: preferences remain static throughout the optimization process, and a priori formalizable objective preferences exist. However, these presuppositions often deviate from practical scenarios: in practice, DM preferences are often dynamic, while non-domain experts may struggle to articulate objective importance through quantitative formulations like weight assignments. Interactive methods [22], [23], while allowing incremental preference feedback, often rely on frequent manual respecification of preferences or on heuristic update rules, which increases the human-computer interaction burden and disrupts the seamless progression of the optimization process. Fur-

thermore, many existing frameworks treat preference learning and optimization as sequential tasks—leading to suboptimal exploration–exploitation tradeoffs. This highlights a broader challenge across interactive AI systems: the need to create a seamless, adaptive feedback loop between the human and the model. Successfully implementing such loops has been key to advancing other fields [24]. These shortcomings collectively highlight a critical gap: the lack of a unified framework that can dynamically infer DM preferences during optimization, seamlessly guide the search toward preference-aligned regions, and rigorously balance exploration with exploitation under tight evaluation budgets.

To bridge this gap, we propose the preference-aware Bayesian optimization (PABO) framework, a unified approach that embeds preference learning directly into the MOO loop. In PABO, each expensive black-box objective is modeled by an independent Gaussian process (GP) surrogate, and candidate solutions are generated in a way that takes into account dynamically varying weight vectors to bias the search toward high-utility regions of the Pareto front. Implicit DM preferences are inferred online through pairwise comparison queries: a Bayesian inference module employs the standard Markov chain Monte Carlo (MCMC) sampling to approximate the posterior distribution over latent utility-weight parameters, eliminating the need for a priori weight specification. Furthermore, a sliding-window memory bank handles nonstationary preferences, enabling real-time adaptation to evolving priorities—such as shifting emphasis from cost-driven to efficacy-driven criteria. To efficiently balance exploration and exploitation under tight evaluation budgets, we propose a self-switching acquisition strategy that monitors a sliding window of recent utility improvements and switches the sampling strategy when utility plateaus. Experiments on synthetic benchmarks and real-world applications demonstrate that PABO achieves competitive solution quality with significantly fewer function evaluations compared to state-of-the-art methods.

The main contributions of this work are threefold.

- 1) Developing a novel PABO framework tailored for multiobjective problems. PABO efficiently integrates DM feedback into the candidate solution generation process so that instead of exhaustively exploring the entire Pareto front, the optimization focuses on regions that match the DM’s preferences.
- 2) Proposing a self-switching acquisition strategy that continuously monitors a window of recent utility improvements to detect stagnation, enabling the algorithm to seamlessly transition between global exploration and focused exploitation.
- 3) Extensive experimental evaluations on both synthetic benchmarks and real-world problems show that the proposed method consistently outperforms state-of-the-art approaches. It achieves faster convergence rates with significantly fewer function evaluations while ensuring higher solution quality aligned with DM preferences.

The remainder of this article is organized as follows. Section II reviews related work on multiobjective BO and preference integration. Section III formalizes the problem setting.

Section IV presents the PABO framework in detail. Section V reports experimental results, and Section VI concludes with discussions and future research directions.

## II. RELATED WORKS

Most multiobjective BO approaches aim at approximating the whole Pareto front. For instance, PESMO [25] chooses the evaluation points to maximally reduce the entropy of the posterior distribution over the Pareto set. TSEMO [26] optimizes approximate GP function samples using NSGA-II and uses a hypervolume-based objective for selecting a batch of points from the population. Spartan BO [9] combines a global GP kernel with multiple movable local kernels to adaptively focus on the most promising regions—thus enhancing local exploitation while preserving global exploration and handling nonstationarity. USEMO [27] generates a set of promising candidates by solving a cheap MOO problem using acquisition functions for each objective, then selects the most uncertain candidate from this Pareto set. Expected improvement (EI) in Pareto hypervolume [28], [29], [30] and sequence uncertainty reduction [31] are two similar approaches. It is important to note that none of these methods support preference specification for the output target under the implicit assumption that all regions are equally valuable, and explore the Pareto front uniformly.

Existing studies attempt to integrate DM preferences through explicit articulation frameworks, which typically manifest as two distinct paradigms: a priori fixed preferences and interactive dynamic preferences. The former requires predefined preference specifications before optimization begins, such as weight vector-based methods [18], [19], region-of-interest (ROI)-based methods [20], [21], methods based on transforming objective thresholds into desirability functions [32], and methods based on the importance ranking of objectives [33], [34]. The latter paradigm enables dynamic preference adjustment through human-in-the-loop interaction. Deb and Kumar [23] achieved preference-driven interactive optimization through the integration of reference directions and EMO. The interactive version of ParEGO proposed by Hakanen and Knowles [22] assumes that preferences are provided for each target in the form of a preferred range. Li et al. [35] proposed a nonuniform mapping scheme that shifts the originally uniformly distributed reference points on a canonical simplex toward an aspiration-level vector specified by the DM and allows the DM to iteratively refine this vector based on intermediate results, thereby automatically steering the search toward the region of interest while still preserving global coverage of the Pareto-front boundary solutions. Although PAC-MOO [36] allows progressive priority weighting via its acquisition decomposition, the requirement for continuous preference respecification persists. Rather than modeling or estimating the true preferences, these approaches require the DM to explicitly specify precise preferences, even though this is often challenging.

In contrast to the aforementioned methods that rely on explicit preference specifications by the DM, recently developed implicit preference modeling approaches infer latent

preferences through more natural and less demanding interactive feedback mechanisms, thereby significantly alleviating the burden of continuous human intervention [37]. These methods, such as BO frameworks incorporating pairwise comparisons or desirability ratings [38], [39], operate by iteratively querying the DM for comparative feedback on presented alternatives. A probabilistic model is then updated based on this feedback to learn an underlying utility function or preference distribution over the objective space. This learned model guides the generation of new candidate solutions that are predicted to align better with the DM's implicitly expressed preferences. While this paradigm offers a powerful and user-centric alternative, eliminating the need for precise numerical specifications and potentially capturing more nuanced or evolving preferences, it currently faces a critical limitation. Unlike our approach, these methods typically lack explicit mechanisms to detect and robustly handle abrupt preference changes during the ongoing optimization process. The learned model may become biased toward earlier feedback, struggling to adapt quickly if the DM's priorities shift significantly mid-search without explicit notification or a dedicated model reset mechanism.

Sampling strategies play a pivotal role in balancing exploration–exploitation tradeoffs during optimization. At the core of BO frameworks, acquisition functions guide the next round of evaluation by quantifying the utility of the candidate solution. Recent advances in preference-aware sampling for multiobjective BO have sought to steer the search toward regions of greatest interest to the DM. For example, Feliot et al. [40] introduced the expected weighted hypervolume improvement (EWHI) criterion, which embeds a user-specified weight distribution into the hypervolume improvement calculation and employs sequential Monte Carlo to efficiently approximate the resulting integral, thereby favoring sampling in regions deemed most important by the DM. Blanchard and Sapsis [41] have shown that treating output regions of high interest as “rare events” and applying likelihood-ratio weighting within the acquisition computation can concentrate sampling on those critical regions without sacrificing scalability in higher dimensions. Astudillo and Frazier [42] proposed an interactive multiattribute BO framework in which candidate solutions are evaluated not only for their worst-case performance across a distribution of plausible utility functions, but are also presented in small “menus” that allow the DM to express feedback with minimal cognitive load, thus balancing exploration, exploitation, and preference elicitation. More recently, Ozaki et al. [39] extended this idea by jointly learning a model of the DM's latent utility from pairwise comparisons and incorporating its uncertainty into the acquisition function; similarly, Ip et al. [43] (PUB-MOBO) learn preferences from pairwise comparisons but further combine the learned utility with a preference-dominated utility function and a local multigradient refinement to ensure proximity to the Pareto front.

### III. PROBLEM SETTING

#### A. MOO With Preferences

In MOO, a solution is said to be Pareto-optimal if no objective can be improved without worsening at least one

other objective. The set of all Pareto-optimal solutions is called the Pareto-optimal set, and its image in the objective space is referred to as the Pareto front. In MOO with preferences, the goal is to balance conflicting objectives while incorporating the DM preferences to prioritize solutions that align with practical priorities. Formally, consider an MOO problem with  $L \geq 2$  objectives

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbf{f}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_L(\mathbf{x})]^T \quad (1)$$

where  $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d$  is the decision variable in a  $d$ -dimensional search space, and  $\mathbf{f} : \mathcal{X} \rightarrow \mathbb{R}^L$  represents  $L$  expensive-to-evaluate black-box objective functions.

Approximating the entire Pareto front becomes computationally prohibitive when evaluations are costly. Instead, preference-aware MOO focuses on identifying a single preferred solution that maximizes the DM's latent utility function  $U$ , where  $U$  implicitly encodes the DM's tradeoffs among objectives. The optimization problem therefore reduces to

$$\mathbf{x}^* = \arg \max_{\mathbf{x} \in \mathcal{X}} U(\mathbf{f}(\mathbf{x})). \quad (2)$$

To make the relation to regret-based evaluation explicit, let the true optimal utility be

$$U^* = \max_{\mathbf{x} \in \mathcal{X}} U(\mathbf{f}(\mathbf{x})) \quad (3)$$

and define the utility regret of an alternative  $\mathbf{x}$  as

$$r(\mathbf{x}) = U^* - U(\mathbf{f}(\mathbf{x})). \quad (4)$$

Minimizing the regret  $r(\mathbf{x})$  is equivalent to maximizing the utility  $U(\mathbf{f}(\mathbf{x}))$ . Hence, the preferred solution may be written equivalently as

$$\mathbf{x}^* = \arg \max_{\mathbf{x} \in \mathcal{X}} U(\mathbf{f}(\mathbf{x})) = \arg \min_{\mathbf{x} \in \mathcal{X}} r(\mathbf{x}). \quad (5)$$

#### B. DM Preferences

The DM seeks a solution that aligns with their subjective priorities. To reduce feedback burden, preferences are elicited via pairwise comparisons of objective values—a cognitively lightweight and practical mechanism. At each iteration  $t$ , the DM is presented with two candidate objective vectors  $\mathbf{f}(\mathbf{x}_a^{(t)})$  and  $\mathbf{f}(\mathbf{x}_b^{(t)})$ , and expresses a preference  $\mathbf{f}(\mathbf{x}_a^{(t)}) > \mathbf{f}(\mathbf{x}_b^{(t)})$  if they favor the outcome  $\mathbf{f}(\mathbf{x}_a^{(t)})$  over  $\mathbf{f}(\mathbf{x}_b^{(t)})$  in terms of utility.

The DM latent utility function  $U : \mathbb{R}^L \rightarrow \mathbb{R}$  governs their choices. Observed preferences are generated by comparing noisy evaluations of  $U$ . Formally, for any pair of objective vectors  $\mathbf{f}(\mathbf{x}_a)$  and  $\mathbf{f}(\mathbf{x}_b)$ , the preference satisfies

$$\mathbf{f}(\mathbf{x}_a) > \mathbf{f}(\mathbf{x}_b) \iff U(\mathbf{f}(\mathbf{x}_a)) + \epsilon_a > U(\mathbf{f}(\mathbf{x}_b)) + \epsilon_b \quad (6)$$

where the notation  $\epsilon_a, \epsilon_b \sim \mathcal{N}(0, \sigma^2)$  indicates that the noise terms are independently and identically distributed according to a Gaussian distribution with a mean of zero and a variance of  $\sigma^2$ . This probabilistic model reflects real-world scenarios where preferences may be inconsistent due to cognitive biases or noisy feedback.

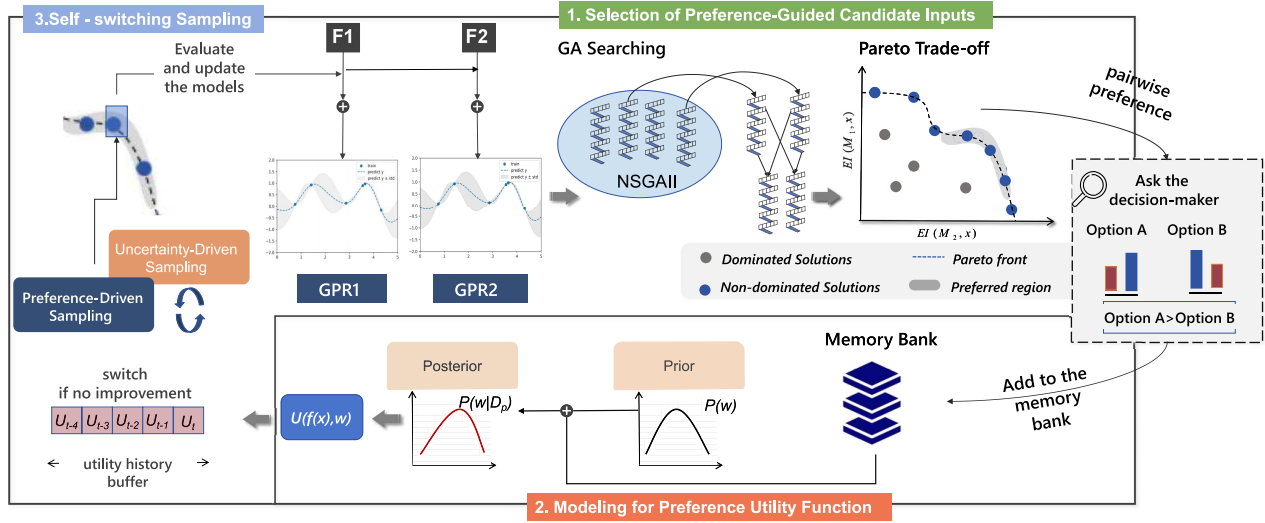


Fig. 1. Overview of the PABO framework for two objective functions. The multiple expensive black-box functions are first modeled using independent GP surrogates. Candidate solutions are generated by applying NSGA-II with a weighted Euclidean distance sorting mechanism to bias the search toward regions aligned with DM preferences. DM feedback is incorporated through pairwise comparisons that are stored in a memory bank and used to update weight vector  $w$ . The updated preferences guide the candidate selection, while a self-switching acquisition strategy dynamically alternates between global exploration and local exploitation.

#### IV. PROPOSED APPROACH

We describe our PABO method as follows. Section IV-A provides an overview of the approach. Section IV-B presents how candidate solutions are generated with DM preferences. Section IV-C details the Bayesian preference model for the utility function. Section IV-D introduces the self-switching acquisition strategy.

##### A. The PABO Algorithm

The workflow of the PABO framework is illustrated in Fig. 1 and summarized in Algorithm 1. The algorithm first builds several GP models based on the initial dataset to probabilistically model each objective function and integrates the DM's preferences into the candidate solution generation process. Unlike methods that strive to approximate the entire Pareto front, it focuses more on the regions of most interest to the DM, thereby effectively reducing resource waste in less relevant areas. Additionally, the algorithm incorporates an adaptive sampling mechanism to balance global exploration and local exploitation, ensuring thorough exploration under high uncertainty and focused optimization under low uncertainty, even under tight evaluation budgets.

By iteratively refining GP models, updating preference weights, and adaptively switching sampling strategies, PABO efficiently converges to solutions that maximize the DM's latent utility while minimizing costly function evaluations.

##### B. Selection of Preference-Guided Candidate Inputs

We construct  $L$  independent Gaussian process regression (GPR) models as surrogates for the multiobjective function. Each acquisition function  $AF(M_i, x)$ —specifically the EI criterion in our implementation—encapsulates the utility of evaluating candidate  $x$  for optimizing its associated objective

##### Algorithm 1 PABO Framework

```

1 Input: initial data, objectives  $\{F_i\}_{i=1}^L$ , sampling limit  $T_{\max}$ 
2 Initialize: Build GP models  $\{M_i\}_{i=1}^L$  from initial data
2 Initialize: utility history buffer  $\mathcal{W} = \emptyset$ 
3 for  $t = 1$  to  $T_{\max}$  do
4   Compute  $EI(x, M_i)$ 
5   Generate candidate set  $\mathcal{X}_p$ 
6   Modeling for Utility Function
7   if Uncertainty-Driven Sampling then
8      $x^* = \arg \max_{x \in \mathcal{X}_p} V(x)$  (Equation 21)
9   else
10     $x^* = \arg \max_{x \in \mathcal{X}_p} U(x)$  (Equation 22)
11   end if
12   Evaluate  $x^*$  to obtain  $\{F_i(x^*)\}_{i=1}^L$ 
13   Update data:  $D \leftarrow D \cup \{(x^*, \{F_i(x^*)\}_{i=1}^L)\}$ 
14   Update GP models
15   Compute current utility  $U_{\text{current}} = U(f(x^*), \mathbf{w})$ 
16   Update  $\mathcal{W}$ : Append  $U_{\text{current}}$  and retain last  $N$  values
17   if no improvement in  $\mathcal{W}$  for  $N$  iterations then
18     // Toggle exploration/exploitation
19     Switch sampling strategy
20   end if
21 end for

```

$F_i$ . Formally, the multiacquisition optimization problem is defined as

$$\max_{x \in X} [EI(M_1, x), EI(M_2, x), \dots, EI(M_L, x)]. \quad (7)$$

To solve this problem, we adapt the NSGA-II algorithm with a weighted Euclidean distance mechanism for nondominated sorting [44]. The distance metric incorporates dynamically adjusted preference weights  $\{w_i\}$  to bias candidate generation



toward DM-preferred regions

$$\text{distance} = \sqrt{\sum_{i=1}^L w_i \left( \frac{f_i^{(k+1)} - f_i^{(k-1)}}{f_i^{\max} - f_i^{\min}} \right)^2} \quad (8)$$

where  $w_i$  denotes the DM's priority weight for objective  $F_i$ ;  $f_i^{\max}$  and  $f_i^{\min}$  represent the maximum and minimum values of  $F_i$  in the current population, respectively; and  $f_i^{(k-1)}$  and  $f_i^{(k+1)}$  correspond to the objective values of the solutions adjacent to solution  $k$  along the Pareto front in the current population. The denominator serves to normalize objectives and mitigate scale disparities. The weights  $w_i$  are iteratively updated via Bayesian inference using DM feedback (Section IV-C), ensuring the candidate set  $\mathcal{C}$  evolves to prioritize regions of higher utility.

By dynamically adjusting the weights  $w_i$ , this approach generates a solution distribution biased toward regions of the Pareto front that align with DM preferences.

### C. Modeling for Utility Function

Traditional scalarization methods like weighted sum scalarization [see (9)] are limited by convexity constraints in Pareto front coverage

$$U(f(x)) = \sum_{i=1}^L w_i f_i(x). \quad (9)$$

To address this limitation, we adopt the weighted Chebyshev scalarization (WCS) which proposed by Gass and Saaty [45], is formally defined as

$$U(f(x)) = \min \left( \frac{f_1(x)}{w_1}, \dots, \frac{f_L(x)}{w_L} \right) \quad (10)$$

where  $w_i$  denotes the preference weight for the  $i$ -th objective, and the ideal point is implicitly defined as the origin due to normalization.

As illustrated in Fig. 2, WCS covers Pareto solutions in nonconvex regions by minimizing the maximum weighted distance between objectives and the ideal point. Theoretical proofs show that when weight vectors are uniformly distributed over the entire weight space, WCS can generate all efficient solutions regardless of the convexity of the Pareto front [46].

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#### Algorithm 2 Modeling for Utility Function

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- 1 **Input:** candidate pool  $\mathcal{X}_p$ , Dirichlet prior parameter  $\alpha$
  - 2 Initialize memory bank  $\mathcal{M} \leftarrow \emptyset$ , prior  $p(\mathbf{w}) \sim \text{Dir}(\alpha)$
  - 3 Select most informative pair  $(y_1, y_2)$
  - 4 Query DM: obtain preference  $r \in \{0, 1\}$  between  $y_1, y_2$
  - 5 Update memory:  $\mathcal{M} \leftarrow \mathcal{M} \cup \{(y_1, y_2, r)\}$
  - 6 Bayesian update of weight posterior
  - 7 **Output:** posterior distribution  $p(\mathbf{w}|\mathcal{D})$
- 

The utility learning algorithm is based on the Bayesian inference framework. It uses MCMC sampling techniques to dynamically adjust the weight vector, approximating the DM implicit preference distribution.

1) *Pairwise Preference Selection:* Our preference elicitation framework transcends conventional approaches by strategically selecting query pairs not merely from evaluated solutions, but rather focusing on promising candidates that exhibit high informational value yet remain unexplored. Formally, at each iteration  $t$ , we identify the maximally informative preference pair through information-theoretic criteria proposed by Houlsby et al. [47]

$$I(r; w|y_1, y_2, \mathcal{D}_t) = H[r|y_1, y_2, \mathcal{D}_t] - \mathbb{E}_w[H[r|w, y_1, y_2]] \quad (11)$$

where  $I$  denotes the mutual information between candidate solutions  $y_1, y_2 \in \mathbb{R}^L \times \mathbb{R}^L$  conditioned on observed data  $\mathcal{D}_t$ , with  $w$  representing the model parameters sampled from the posterior  $p(w|\mathcal{D}_t)$ ,  $r(y_1, y_2) \in \{0, 1\}$  being the binary preference feedback. The conditional entropy term within this expression measures the uncertainty in preference predictions for fixed model parameters

$$H[r|w, y_1, y_2] = - \sum_{r \in \{0, 1\}} P(r|w, y_1, y_2) \log P(r|w, y_1, y_2). \quad (12)$$

For practical computation, Monte Carlo approximations are employed using posterior samples  $w$ . The predictive distribution of preferences is approximated via posterior averaging

$$p(r|y_1, y_2, \mathcal{D}_t) \approx \frac{1}{|\mathcal{W}|} \sum_{w \in \mathcal{W}} p(r|w, y_1, y_2). \quad (13)$$

Similarly, the expected conditional entropy term is approximated by averaging entropies across parameter samples

$$\begin{aligned} \mathbb{E}_w[H[r|w, y_1, y_2]] \\ \approx \frac{1}{|\mathcal{W}|} \sum_{w \in \mathcal{W}} \sum_{r \in \{0, 1\}} p(r|w, y_1, y_2) \log p(r|w, y_1, y_2). \end{aligned} \quad (14)$$

The optimal query pair that maximizes information gain about model parameters is then selected though

$$(y_1^*, y_2^*) = \arg \max_{(y_1, y_2) \in \mathcal{X}_p} I(r; w|y_1, y_2, \mathcal{D}_t) \quad (15)$$

where  $\mathcal{X}_p$  represents the candidate pool containing unexplored solutions with high uncertainty measures. Upon selecting the optimal pair, we observe the DM comparative feedback  $r(y_1^*, y_2^*)$  and update the probabilistic model accordingly.

2) *Memory Bank Mechanism:* To address nonstationary preferences in dynamic environments, the algorithm introduces a sliding window memory bank mechanism.

This mechanism progressively historical information by replacing old samples, avoiding interference from outdated preferences in current decisions while keeping computational complexity manageable. The preference acquisition process is conducted through solution vectors in the objective space, where the preference pairs stored in the memory repository are utilized for joint likelihood computation. We employ the probit model originating from Thurstone–Mosteller's law of comparative judgment [48] to process comparative data, which establishes a connection between observed preference pair data and continuous latent functions through an underlying

decision-making mechanism. For a single preference pair, its likelihood is defined as

$$P(f_i > f'_i | w) = \Phi \left( \frac{U(f_i, w) - U(f'_i, w)}{\sqrt{2}\sigma} \right) \quad (16)$$

where  $\Phi$  is the cumulative distribution function of the standard Gaussian distribution.

The joint likelihood is calculated by the product of independent probabilities of each pair of preferences in the log space, enabling the algorithm to handle batch preference data.

3) *Prior and Posterior*: Assume the weight vector  $\mathbf{w}$  follows a Dirichlet distribution with parameter  $\alpha$ , where each  $\alpha_i$  is initially set to 2.0 to maintain a uniform expectation ( $\mathbb{E}[w_i] = 1/K$ ) over  $K$  objectives while actively suppressing extreme weight assignments (e.g.,  $w_i \approx 0$  or  $w_i \approx 1$ )

$$p(w) \sim \text{Dir}(\alpha), \quad \alpha = [\alpha_1, \alpha_2, \dots, \alpha_L]. \quad (17)$$

Adjusting the asymmetry of  $\alpha$  can incorporate domain prior knowledge (such as historical preferences), but this article defaults to maintaining symmetry.

After observing a set of preference pairs  $\mathcal{D}$ , the posterior distribution of weights is updated using Bayes' theorem

$$p(w|\mathcal{D}) \propto p(\mathcal{M}|w) p(w) \quad (18)$$

where  $\mathcal{M}$  represents the memory bank of preference pairs. Due to the nonconjugacy of the likelihood function (caused by the Probit model making the Dirichlet prior intractable for posterior inference), the algorithm uses MCMC sampling to approximate the posterior distribution.

#### D. Self-Switching Acquisition Algorithm

To reconcile the conflicting demands of global exploration and local exploitation, we propose a performance-driven window criterion for self-switching acquisition. The mechanism monitors the optimization progress through a sliding window of historical utility values and triggers strategy transitions when stagnation is detected. In each iteration  $t$ , let  $U_{\text{best}}(t)$  denote the highest utility observed so far, computed via the preference-aligned utility function  $U(f(x), \mathbf{w})$ . The system maintains a window  $\mathcal{W}$  storing the improvement status of the most recent  $N = 5$  iterations. At each iteration  $t$ , we record whether  $U_{\text{best}}^{(t)} > U_{\text{best}}^{(t-1)}$ . If no improvement occurs within the window  $\mathcal{W}$ , the algorithm switches its sampling strategy. The two strategies we take are as follows.

1) *Uncertainty-Driven Sampling*: When  $U_{\text{best}}$  remains unchanged for  $N$  consecutive iterations, the system assumes local exploitation has saturated the current region's potential and activates global exploration to escape local optima. Uncertainty-driven sampling prioritizes regions with high predictive variance

$$\text{UB}_i(\mathbf{x}) = \mu_i(\mathbf{x}) + \beta\sigma_i(\mathbf{x}) \quad (19)$$

$$\text{LB}_i(\mathbf{x}) = \mu_i(\mathbf{x}) - \beta\sigma_i(\mathbf{x}) \quad (20)$$

$$V(\mathbf{x}) = \prod_{i=1}^M (\text{UB}_i(\mathbf{x}) - \text{LB}_i(\mathbf{x})). \quad (21)$$

$\text{LB}(\mathbf{x})$  and  $\text{UB}(\mathbf{x})$  denote the lower and upper confidence bounds of the statistical model at input  $\mathbf{x}$ , respectively. The time-varying parameter  $\beta_t$  governs the tradeoff between exploitation and exploration at iteration  $t$ , following the adaptive scheduling strategy proposed by Srinivas et al. [49]. This theoretical construction guarantees sublinear growth of cumulative regret in stochastic multiarmed bandit settings. Our acquisition criterion measures the uncertainty volume metric for all candidate inputs  $\mathbf{x} \in \mathcal{X}_p$  and selects the input with maximal uncertainty measure for subsequent function evaluation.

2) *Preference-Driven Sampling*: When the algorithm shifts its focus from global exploration to local exploitation, the acquisition strategy leverages the preference information provided by the objective functions. Specifically, we adopt a scalarization method to compute a preference score  $\phi(\mathbf{x})$  for each candidate solution  $\mathbf{x} \in \mathcal{X}_p$ , which aggregates multiple objectives into a single scalar metric reflecting the DM preferences. Formally, letting  $\phi(\mathbf{x})$  denote the composite preference score, we select the candidate solution with the highest preference score, as the input for subsequent evaluation

$$\mathbf{x}^* = \arg \max_{\mathbf{x} \in \mathcal{X}_p} \phi(\mathbf{x}). \quad (22)$$

In practice, the preference score can be computed by integrating local gradient information of the objectives or by directly evaluating the utility function. In our proposed framework, we adopt the latter approach: the preference score is computed by directly evaluating the utility function defined in 10 using the currently learned weight vector obtained through Bayesian inference. This mode is particularly effective in regions where model uncertainty is low, as it focuses on fine-tuning within promising areas identified during the exploration phase.

The switching mechanism achieves a dynamic balance between global exploration and preference-driven optimization by adaptively responding to the optimization progress. It makes efficient decisions in dynamic multiobjective environments by monitoring historical utility patterns and triggering strategy switching when stagnation is detected. The principle of the mechanism is as follows.

- 1) *Avoiding Local Optima Stagnation*: Persistent utility stagnation signals diminishing returns from local exploitation, necessitating exploration to discover unexplored regions.
- 2) *Computational Efficiency*: The windowed approach prevents premature switching caused by transient performance fluctuations, ensuring stable strategy execution.
- 3) *Dynamic Adaptation*: The mechanism autonomously responds to optimization phases without manual threshold tuning, enhancing robustness across diverse problems.

## V. EXPERIMENTAL RESULTS

The proposed framework<sup>1</sup> is validated through a structured experimental analysis spanning progressive phases: preference

<sup>1</sup><https://github.com/kg4sci/PABO>

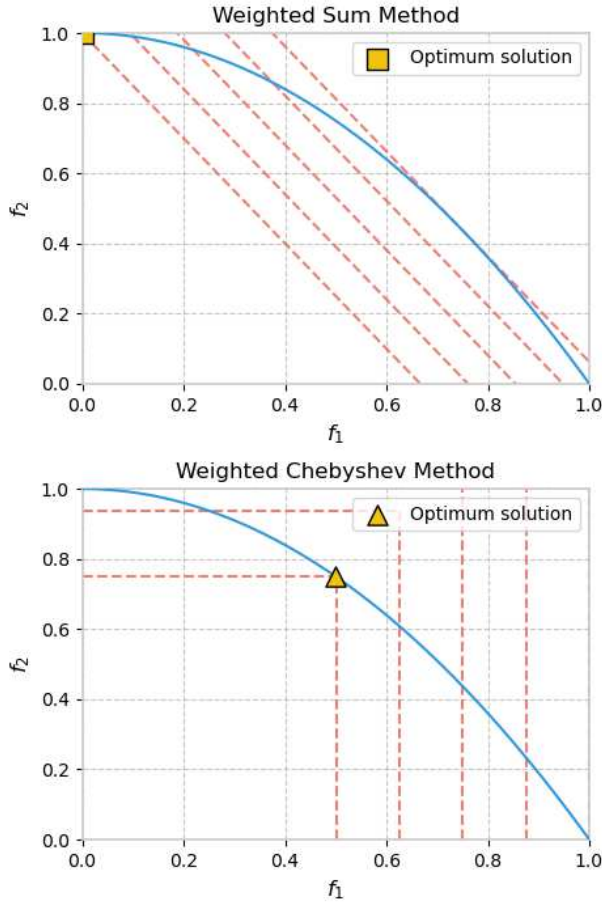


Fig. 2. WS versus WCS performance on nonconvex Pareto front: WS misses nonconvex solutions; WCS captures entire front.

learning accuracy, dynamic preference adaptation, comparative performance benchmarking, and componentwise ablation studies.

#### A. Error Assessment for Preference Learning

We iteratively introduce preference information to evaluate the estimation accuracy of the parameter  $w$ . In each iteration, a preference pair is selected and added to the memory bank, which is then used to update the posterior distribution and re-estimate  $w$ .

Fig. 3 illustrates the experimental results. The horizontal axis corresponds to the iteration, and the vertical axis shows error  $= \|w - w_{\text{true}}\|_2$ . The results are averaged over ten independent experiments. As shown, both in the case of high-dimensional and low-dimensional parameter space, the estimation error quickly drops to approximately 0.1 within only a few dozen iterations.

Since preference-guided optimization tasks do not require extremely high precision in parameter estimation—and in practical scenarios, a error level of about 0.1 is often sufficient to ensure fast convergence to a satisfactory solution—we conclude that the achieved estimation precision meets the requirements for preference-guided optimization. In subse-

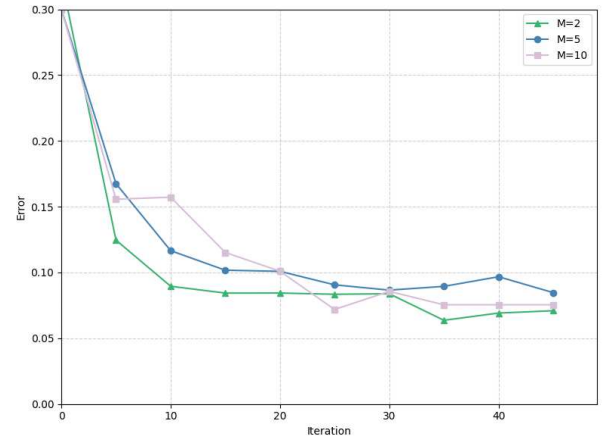


Fig. 3. Estimation error of  $w$  across dimensions ( $M = 2, 5, 10$ , where  $M$  denotes the number of objectives).

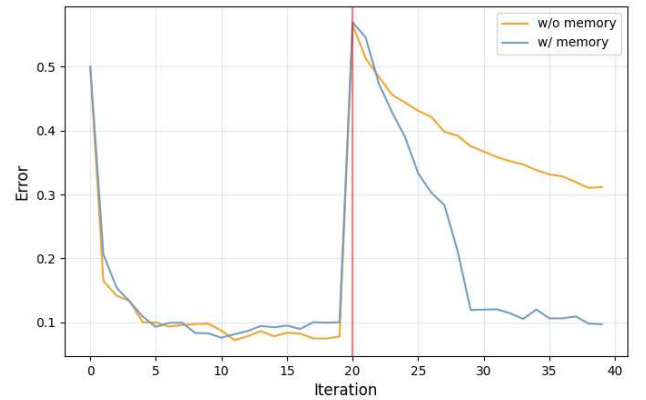


Fig. 4. Impact of memory bank mechanism on dynamic preference adaptation.

quent experiments, we also demonstrated that  $w$  contains adequate informative guidance to drive efficient exploration.

To evaluate PABO's adaptability to dynamic preferences, we simulate abrupt preference shifts by resampling  $w_{\text{true}}$  from a Dirichlet distribution.

As illustrated in Fig. 4, when an abrupt preference shift occurs ( $t = 20$ ), PABO's weight estimation error initially peaks but sharply decreases and stabilizes within a short period, owing to the memory bank mechanism discarding outdated preferences. In contrast, baselines without this mechanism exhibit persistently high errors due to historical data interference, leading to delayed alignment with new preferences. On the other hand, in the scenario with fixed preferences, this mechanism can prevent the model update speed from gradually slowing down in later iterations due to the increase in preference pairs.

#### B. Optimize Performance Comparisons

1) *Baseline*: Based on differing objectives, we group our baselines into two categories. First, global Pareto-front approximators such as ParEGO [50], USeMO [27], and PDBO [51] aim to recover as much of the entire trade-off surface as possible; we compare against them to quantify

TABLE I

DETAILS OF SYNTHETIC BENCHMARKS: NAME, BENCHMARK FUNCTIONS, NO. OF OBJECTIVES  $M$ , AND INPUT DIMENSION  $d$

| Name       | Benchmark functions                             | $M$ | $d$ |
|------------|-------------------------------------------------|-----|-----|
| DTLZ1      | Deb, Thiele, Laumanns, Zitzler [53]             | 2   | 6   |
| ZDT1       | Zitzler, Deb, Thiele [54]                       | 2   | 4   |
| PRDZPS-6-6 | Powell-Rastrigin-Dixon-Zakharov-Perm-SumSquares | 6   | 6   |

how well PABO’s targeted search—despite not aiming for full-front coverage—identifies high-quality solutions relative to exhaustive-coverage methods. Second, preference-guided methods such as EIUU [42] and BOPE [52] focus their budget on regions most likely to satisfy DM preferences; by contrasting PABO with these, we demonstrate how our algorithm’s explicit incorporation of pairwise preference feedback and adaptive exploration–exploitation balancing leads to faster convergence toward truly preferred optima.

To establish an upper bound benchmark, we also include an “Ours with true preference” variant, which assumes perfect knowledge of the DM’s true preference weights from the outset.

2) *Implementation Details:* We use a GP-based statistical model with squared exponential (SE) kernel in all our experiments. The GP model is initialized using  $2(d + 1)$  points sampled from the design space (where  $d$  denotes the dimensionality), and objective function values are normalized to ensure all objectives share a common scale.

The true preference parameter  $w_{\text{true}}$  is determined by sampling from the Dirichlet distribution, ensuring that  $\sum_{i=1}^L w_i = 1$ . We evaluate the efficiency of the proposed method by computing the simple regret with respect to the true utility function  $U_{w_{\text{true}}}$ . The simple regret  $R$  evaluations is defined as

$$R = U^* - \max_{x \in \Theta_s} U_{w_{\text{true}}}(f(x)). \quad (23)$$

Here,  $U^*$  denotes the maximum attainable utility, computed in advance by uniformly sampling 5000 points in the design space and selecting the highest utility value observed, which serves as the baseline for regret. The set  $\Theta_s$  comprises all points evaluated during the experiment. During each experimental round, the algorithm selects a candidate point for objective function evaluation, and the DM provides preference information through pairwise comparisons. To quantify performance consistency, the mean and 90% confidence interval of the results were computed over ten independent runs.

3) *Synthetic Benchmarks:* We validate the proposed method on widely used benchmarks in MOO literature. Additionally, we construct a synthetic MOO benchmark problem by combining commonly employed benchmark functions from single-objective optimization. We provide the details of these MOO benchmarks in Table I.

4) *Real-World Benchmarks:* Most synthetic questions have unusual features (e.g., the triangular Pareto front, where the term “triangular” means that the shape of the Pareto front

TABLE II

DETAILS OF REAL-WORLD BENCHMARKS: NAME, BENCHMARK FUNCTIONS, NO. OF OBJECTIVES  $M$ , AND INPUT DIMENSION  $d$

| Name                    | Benchmark functions             | $M$ | $d$ |
|-------------------------|---------------------------------|-----|-----|
| Car side impact design  | H. Jain and K. Deb [55]         | 3   | 7   |
| Hatch cover design      | H. M. Amir and T. Hasegawa [56] | 2   | 2   |
| Water resource planning | R. Tanabe and H. Ishibuchi [57] | 6   | 3   |
| Rocket injector design  | R. Tanabe and H. Ishibuchi [57] | 3   | 4   |
| Two bar truss design    | R. Tanabe and H. Ishibuchi [57] | 3   | 3   |
| Disc brake design       | R. Tanabe and H. Ishibuchi [57] | 3   | 4   |

looks like a distorted/bended triangle in a three-dimensional objective space [58].) that are unlikely to appear in real-world applications. These artificial characteristics may lead to overestimation of algorithm performance in idealized scenarios, while failing to reflect practical challenges such as high-dimensional objective conflicts, irregular Pareto front geometries, and complex constraints. Therefore, optimization algorithms should also be benchmarked on practical problems to yield more reliable evaluations. In our experiments, we employed six diverse real-world benchmarks; the details are provided in Table II.

5) *Results:* The results are shown in Fig. 5. PABO significantly reduces required evaluations while achieving comparable or better solution quality than state-of-the-art methods. For black-box optimization problems involving expensive function evaluations, a faster convergence enables DM to terminate the optimization process earlier, thereby significantly reducing computational costs and resource consumption without compromising solution quality. Moreover, accelerated convergence facilitates more efficient decision-making in practical applications, enhancing the system’s responsiveness and practical utility. On the other hand, similar performance between the PABO and PABO with true preference demonstrates that our utility learning module rapidly and accurately infers the DM’s underlying utility landscape (within the first 20–30 pairwise queries) providing sufficiently informative guidance to drive exploration.

### C. Ablation Study: Effectiveness of Self-Switching Sampling

Recall that PABO needs to select an input from promising candidates with preference guidance for evaluation. In Fig. 6, we compare the self-switching sampling strategy employed by PABO with uncertainty-driven sampling, utility-driven sampling, and random selection sampling. For enhanced visualization, as the raw differences in simple regret values were relatively small and difficult to discern clearly on a linear scale, we present the simple regret on a base-2 logarithmic ( $\log_2$ ) scale. It can be observed that the self-switching strategy achieves the best performance, demonstrating that this mechanism can adaptively select optimal sampling strategies based on the characteristics of optimization phases, effectively balancing the exploration–exploitation tradeoff.



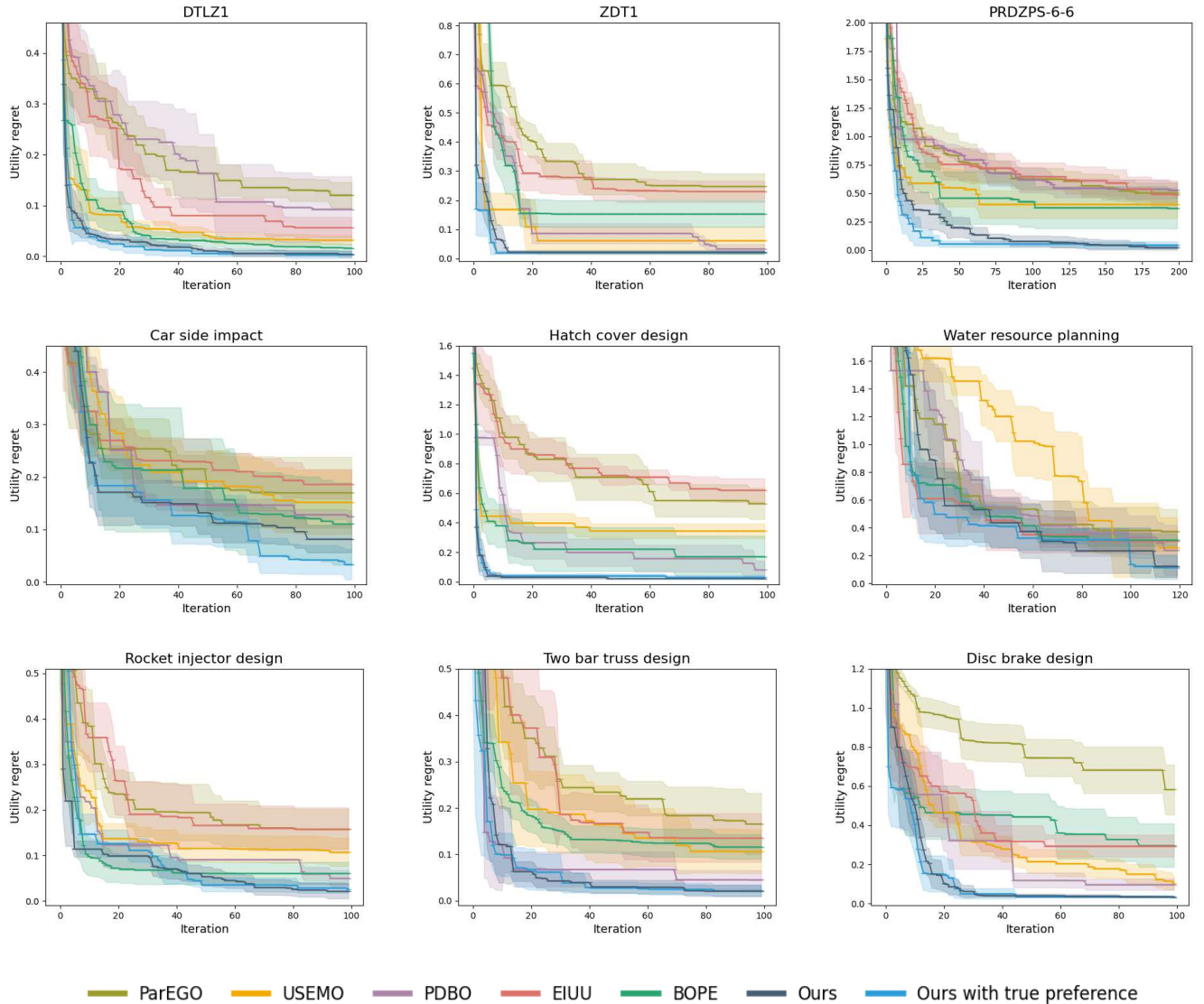


Fig. 5. Performance of different multiobjective BO algorithms on synthetic and real-world benchmarks, averaged over ten independent runs.

A noteworthy observation of ablation studies is that the alternative strategies (uncertainty-driven, utility-driven, and even random) occasionally exhibit performance comparable to the self-switching strategy during specific, localized stages of the optimization. This suggests two key points: first, our underlying method for generating and selecting high-potential candidate solutions is fundamentally sound and effective, as even simpler strategies can achieve good results when applied judiciously. Second, different sampling methods inherently possess local advantages depending on the specific state or region of the problem landscape being explored (e.g., uncertainty-driven might excel in unexplored regions, while utility-driven might be better near promising optima). However, critically, the self-switching strategy's key strength lies in its ability to recognize these contextual shifts and seamlessly switch strategies. Consequently, when considering the entire optimization trajectory from start to finish, the self-switching

strategy demonstrates significantly superior and more robust performance, outperforming any single static strategy overall.

#### D. Parameter Study: Utility Buffer Configuration on Optimization Performance

The selection of the utility buffer window size ( $N = 5$ ) is empirically validated as the optimal configuration, which balances responsiveness, stability, and computational efficiency. This optimal setting can be attributed to the following factors: a smaller window size ( $N < 5$ ) may lead to premature strategy switching due to transient performance fluctuations, while a larger window ( $N > 5$ ) delays the detection of genuine stagnation, prolonging suboptimal exploitation. The choice of  $N = 5$  ensures sufficient tolerance to noise while maintaining timely adaptation to preference shifts.

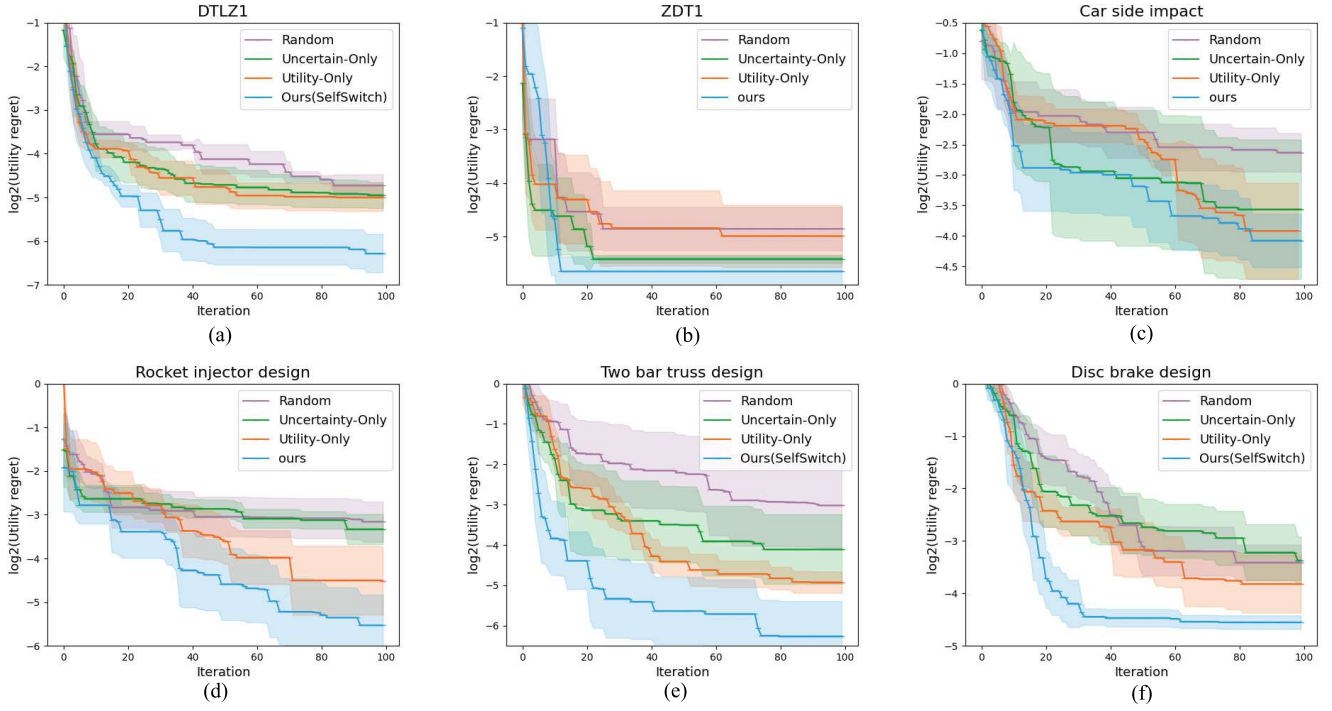


Fig. 6. Comparison of the proposed self-switching acquisition strategy with the uncertainty-driven, utility-driven, and random sampling methods. Results are averaged over ten independent runs: (a) DTLZ1, (b) ZDT1, (c) car side impact, (d) rocket injector design, (e) two bar truss design, and (f) disk brake design.

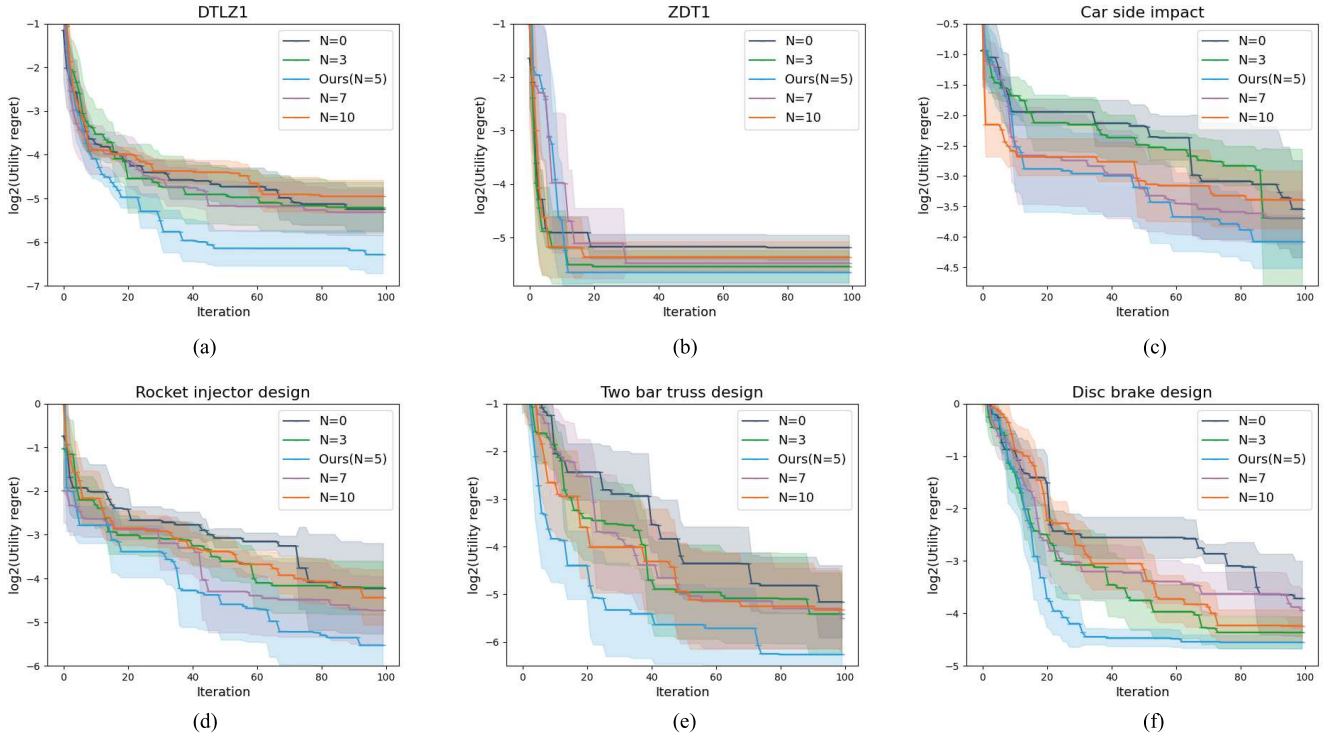


Fig. 7. Comparison of the impact of different utility buffer window sizes ( $N$ ) on optimization performance. Results are averaged over ten independent runs (a) DTLZ1, (b) ZDT1, (c) car side impact, (d) rocket injector design, (e) two bar truss design, and (f) disk brake design.

To rigorously validate this choice, we compare the convergence trajectories across different buffer window sizes ( $N = 0, 3, 5, 7, 10$ ), where  $N = 0$  indicates the absence of any buffer window, and the algorithm only relies on the current utility value for sampling mode switch-

ing. As clearly demonstrated in Fig. 7, the configuration with  $N = 5$  consistently achieves the lowest simple regret across the evaluated finite iterations. This result empirically confirms the superiority of  $N = 5$  in minimizing regret.

## VI. CONCLUSION

In this article, we propose PABO, a novel framework for MOO that dynamically integrates DM preferences into the optimization loop. Unlike traditional methods that either exhaustively approximate the entire Pareto front or rely on static preference assumptions, PABO addresses the critical need for adaptive alignment with evolving DM priorities while maintaining computational efficiency. By embedding preference elicitation through Bayesian inference and introducing a self-switching acquisition strategy, our framework achieves a principled balance between global exploration and local exploitation, even under stringent evaluation budgets. We believe PABO represents a significant step toward practical, human-centric BO in multiobjective settings, enabling on-the-fly optimization within interactive design platforms for engineering design. Looking ahead, we plan to extend the PABO framework along several promising research directions. One important avenue would be to enhance its scalability in high-dimensional objective spaces. Furthermore, introducing mechanisms to detect and correct persistent systemic biases would significantly strengthen PABO's applicability in real-world, nonstationary decision environments. We also intend to extend PABO to collaborative decision-making scenarios with multiple DMs and apply it to a broader range of task types.

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